

# UNIT 2

## AGILE PROJECT DOCUMENTATION

### Project Title

Enterprise Student Record and Academic Management Platform

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### Introduction

The Enterprise Student Record and Academic Management Platform is designed to manage student academic information in a centralized and organized system.

Unit 2 focuses on applying Agile and Scrum methodologies to the development of the platform. The project is divided into smaller deliverable features called Epics and User Stories. These features are prioritized in a Product Backlog and implemented through Sprints.

Agile development enables continuous feedback, flexibility, faster delivery, and continuous improvement throughout the project.

## 1. Agile Product Vision

### Product Vision Statement

To develop a centralized, secure, and user-friendly academic management platform that enables students, faculty, and administrators to efficiently manage student records, courses, attendance, examinations, marks, and academic results.

### Vision Goals

- Centralize student academic information.
- Reduce manual academic record management.
- Provide role-based access to users.
- Improve accuracy of student records.
- Provide faster access to academic information.
- Enable efficient course, attendance, and examination management.
- Support continuous improvement through Agile development.

## 3. Scrum Team Structure

The project follows the Scrum framework.

## Scrum Team

| Role             | Responsibility                                           |
|------------------|----------------------------------------------------------|
| Product Owner    | Defines product requirements and prioritizes the backlog |
| Scrum Master     | Facilitates Scrum activities and removes obstacles       |
| Development Team | Designs, develops, tests, and integrates the system      |

## Stakeholders

- Students
- Faculty Members
- Academic Administrators
- Examination Cell Staff
- IT Support Team

## 2. Scrum Roles and Responsibilities

### Product Owner

- Defines product vision.
- Identifies user requirements.
- Creates and prioritizes the Product Backlog.
- Communicates with stakeholders.
- Accepts or rejects completed features.

### Scrum Master

- Organizes Scrum ceremonies.
- Ensures Scrum practices are followed.
- Removes development blockers.
- Supports communication within the team.
- Helps the team improve continuously.

### Development Team

- Analyzes requirements.

- Designs system components.
- Develops application features.
- Performs testing.
- Fixes defects.
- Integrates completed features.

## 3. Project Epics

The major features of the system are divided into the following Epics.

### Epic 1: User Authentication and Access Control

Provides secure login and role-based access for students, faculty, administrators, and other authorized users.

### Epic 2: Student Record Management

Allows administrators to create, update, search, and maintain student academic records.

### Epic 3: Course Registration

Allows students to view available courses and register for required courses.

### Epic 4: Attendance Management

Allows faculty members to record attendance and students to view their attendance information.

### Epic 5: Examination and Marks Management

Allows examination staff and faculty to manage examinations, marks, and assessments.

### Epic 6: Results and Academic Reports

Provides students and administrators with academic results and reports.

### Epic 7: Role-Based Dashboards

Provides customized dashboards according to the user's role.

## 4. User Stories

### Epic 1: Authentication

US01:

As a student, I want to log in securely so that I can access my academic information.

US02:

As an administrator, I want role-based access so that users can access only authorized features.

## Epic 2: Student Records

US03:

As an administrator, I want to add student records so that student information is maintained digitally.

US04:

As an administrator, I want to update student information so that records remain accurate.

US05:

As an administrator, I want to search student records so that I can quickly find required information.

## Epic 3: Course Registration

US06:

As a student, I want to view available courses so that I can select suitable courses.

US07:

As a student, I want to register for courses so that my academic registration is recorded.

## Epic 4: Attendance

US08:

As a faculty member, I want to record student attendance so that attendance data is maintained.

US09:

As a student, I want to view my attendance so that I can monitor my attendance percentage.

## Epic 5: Examination and Marks

US10:

As examination staff, I want to create examination records so that examinations can be managed systematically.

US11:

As a faculty member, I want to enter student marks so that assessment results can be maintained.

## Epic 6: Results and Reports

US12:

As a student, I want to view my examination results so that I can track my academic

performance.

US13:

As an administrator, I want to generate academic reports so that student performance can be analyzed.

## Epic 7: Dashboards

US14:

As a user, I want a role-based dashboard so that I can quickly access relevant information and functions.

## 5. Acceptance Criteria

| User Story | Acceptance Criteria                                         |
|------------|-------------------------------------------------------------|
| US01       | User can enter valid credentials and access the system      |
| US02       | Users can access only features permitted for their role     |
| US03       | Administrator can add a new student record successfully     |
| US04       | Administrator can edit and save updated student information |
| US05       | Administrator can search students using relevant details    |
| US06       | Student can view available courses                          |
| US07       | Student can select and register for available courses       |
| US08       | Faculty can record attendance for students                  |
| US09       | Student can view current attendance information             |
| US10       | Examination staff can create examination                    |
|            | records                                                     |
| US11       | Faculty can enter and save marks                            |

|      |                                                    |
|------|----------------------------------------------------|
| US12 | Student can view published examination results     |
| US13 | Administrator can generate academic reports        |
| US14 | Dashboard displays features according to user role |

## 6. Product Backlog

| ID   | Product Backlog Item   | Priority | Story Points |
|------|------------------------|----------|--------------|
| US01 | Secure Login           | High     | 3            |
| US02 | Role-Based Access      | High     | 5            |
| US03 | Add Student Record     | High     | 5            |
| US04 | Update Student Record  | High     | 3            |
| US05 | Search Student Records | Medium   | 3            |
| US06 | View Available Courses | High     | 3            |
| US07 | Course Registration    | High     | 5            |
| US08 | Record Attendance      | High     | 5            |
| US09 | View Attendance        | Medium   | 3            |
| US10 | Examination Management | High     | 5            |
| US11 | Marks Management       | High     | 5            |

|      |                      |        |   |
|------|----------------------|--------|---|
| US12 | View Results         | High   | 3 |
| US13 | Academic Reports     | Medium | 5 |
| US14 | Role-Based Dashboard | Medium | 5 |

# 7. Story Point Estimation

Story points are used to estimate the relative complexity and effort required for each User Story.

The project uses the following scale:

1, 2, 3, 5, 8

## Estimation Guidelines

- 1–2 points: Very simple task
- 3 points: Small task with limited complexity
- 5 points: Medium-complexity task
- 8 points: Large or highly complex task Story points

consider:

- Complexity
- Development effort
- Testing effort
- Dependencies
- Technical uncertainty

# 8. Sprint Planning

The project is divided into short development cycles called Sprints. Each Sprint focuses on delivering a working set of features.

## Sprint Duration

2 Weeks per Sprint

## Sprint Objectives

### Sprint 1

Focus on system foundation and authentication. Features:

- Secure Login
- Role-Based Access
- Basic Dashboard

### Sprint 2

Focus on student record and course management. Features:

- Add Student Record
- Update Student Record
- Search Student Records
- View Available Courses
- Course Registration

### Sprint 3

Focus on attendance and examination management. Features:

- Record Attendance
- View Attendance
- Examination Management
- Marks Management

### Sprint 4

Focus on results, reports, testing, and final integration. Features:

- View Results
- Academic Reports
- Final Dashboard Improvements
- System Testing
- Bug Fixing

# 9. Sprint Backlog

## Sprint 1 Backlog

| ID   | Task                       | Story Points |
|------|----------------------------|--------------|
| US01 | Secure Login               | 3            |
| US02 | Role-Based Access          | 5            |
| US14 | Basic Role-Based Dashboard | 5            |

Total: 13 Story Points

## Sprint 2 Backlog

| ID   | Task                   | Story Points |
|------|------------------------|--------------|
| US03 | Add Student Record     | 5            |
| US04 | Update Student Record  | 3            |
| US05 | Search Student Records | 3            |
| US06 | View Available Courses | 3            |
| US07 | Course Registration    | 5            |

Total: 19 Story Points

## Sprint 3 Backlog

| ID   | Task              | Story Points |
|------|-------------------|--------------|
| US08 | Record Attendance | 5            |
| US09 | View Attendance   | 3            |

|      |                        |   |
|------|------------------------|---|
| US10 | Examination Management | 5 |
| US11 | Marks Management       | 5 |

Total: 18 Story Points

## Sprint 4 Backlog

| ID   | Task                          | Story Points |
|------|-------------------------------|--------------|
| US12 | View Results                  | 3            |
| US13 | Academic Reports              | 5            |
| US14 | Dashboard Improvements        | 5            |
| -    | System Testing and Bug Fixing | 5            |

Total: 18 Story Points

# 10. Scrum Board

The Scrum Board is used to track the progress of tasks during a Sprint.

| To Do           | In Progress         | Testing            | Done          |
|-----------------|---------------------|--------------------|---------------|
| Student Records | Course Registration | Login Testing      | Project Setup |
| Attendance      | Dashboard           | Course Testing     | Requirements  |
| Reports         | Marks Management    | Attendance Testing | UI Prototype  |
| Results         | -                   | -                  | -             |

The board helps the team visualize work and identify tasks that are blocked or delayed.

# 11. Burndown Chart

A Burndown Chart represents the amount of remaining work during a Sprint.

## Example Sprint Burndown Data

| Sprint Day | Remaining Story Points |
|------------|------------------------|
| Day 1      | 13                     |
| Day 3      | 11                     |
| Day 5      | 9                      |
| Day 7      | 6                      |
| Day 9      | 3                      |
| Day 10     | 0                      |

The expected trend is a gradual reduction in remaining work until all Sprint tasks are completed.

# 12. Sprint Review

At the end of every Sprint, the completed work is demonstrated to stakeholders.

## Sprint 1 Review

Completed:

- Login functionality
- Role-based access
- Initial dashboard

## Sprint 2 Review

Completed:

- Student record management
- Student search

- Course viewing
- Course registration

## Sprint 3 Review

Completed:

- Attendance management
- Examination management
- Marks entry

## Sprint 4 Review

Completed:

- Results
- Academic reports
- Dashboard improvements
- Final testing and integration

Stakeholder feedback is collected and used to update the Product Backlog.

# 13. Sprint Retrospective

The Sprint Retrospective is conducted after each Sprint to identify improvements.

## What Went Well

- Clear division of tasks.
- Regular communication between team members.
- Incremental feature development.
- Continuous testing of completed features.

## What Could Be Improved

- Improve estimation accuracy.
- Reduce task dependencies.
- Improve documentation consistency.
- Identify technical issues earlier.

## Action Items

- Conduct better Sprint Planning.
- Break large tasks into smaller tasks.
- Perform testing throughout the Sprint.
- Maintain regular communication.
- Review backlog priorities regularly.

# 14. Agile Framework Comparison

|                   |                      |                     |                         |
|-------------------|----------------------|---------------------|-------------------------|
| Feature           | Scrum                | Kanban              | XP                      |
| Work Cycle        | Fixed Sprints        | Continuous          | Iterative               |
| Roles             | Clearly defined      | Flexible            | Defined technical roles |
| Planning          | Sprint Planning      | Continuous          | Frequent planning       |
| Best For          | Product development  | Continuous workflow | Software quality        |
| Progress Tracking | Scrum Board/Burndown | Kanban Board        | Engineering practices   |
| Project Choice    | Selected             | Alternative         | Alternative             |

## Selected Framework

Scrum is selected for this project because the platform can be developed incrementally through defined Sprints with clear roles, backlog management, Sprint Reviews, and Retrospectives.

# 15. Project Timeline and Release Roadmap

| Phase    | Duration | Major Deliverables                      |
|----------|----------|-----------------------------------------|
| Sprint 1 | Week 1–2 | Authentication and Dashboard            |
| Sprint 2 | Week 3–4 | Student Records and Course Registration |
| Sprint 3 | Week 5–6 | Attendance and Examination Management   |
| Sprint 4 | Week 7–8 | Results, Reports, Testing               |
|          |          | and Integration                         |

# Release Roadmap

## Release 1 – Foundation

- Login
- Authentication
- Role-Based Access
- Dashboard

## Release 2 – Academic Management

- Student Records
- Course Registration
- Attendance

## Release 3 – Examination Management

- Examination Management
- Marks Management
- Results

## Release 4 – Final Product

- Academic Reports
- Testing
- Bug Fixing
- Final Integration

# 16. Agile Development Benefits

Agile development provides the following benefits for this project:

- Faster delivery of features.
- Continuous stakeholder feedback.
- Flexible response to changing requirements.
- Early identification of defects.
- Incremental development.
- Better team collaboration.
- Continuous product improvement.

# 17. Final Deliverables

The Unit 2 Agile project documentation includes:

- ✓ Agile Product Vision
- ✓ Scrum Team Structure
- ✓ Scrum Roles and Responsibilities
- ✓ Project Epics
- ✓ User Stories
- ✓ Acceptance Criteria
- ✓ Product Backlog
- ✓ Story Point Estimation
- ✓ Sprint Planning
- ✓ Sprint Backlog
- ✓ Scrum Board
- ✓ Burndown Chart
- ✓ Sprint Review
- ✓ Sprint Retrospective
- ✓ Agile Framework Comparison
- ✓ Project Timeline and Release Roadmap

## 18. Conclusion

The Agile and Scrum methodology provides a structured approach for developing the Enterprise Student Record and Academic Management Platform.

By dividing the project into Epics, User Stories, Product Backlog items, and Sprints, the development team can deliver the platform incrementally while continuously incorporating stakeholder feedback.

The Scrum framework also promotes collaboration, transparency, continuous testing, and improvement, making it suitable for the development of this academic management platform.